GHOST INFRASTRUCTURE — Development Log

This is my personal coding diary for the Ghost Infrastructure project. I am trying to see if the old 19th and 20th-century coal mine maps of Bochum still secretly control who gets a nice, walkable "15-minute neighborhood life" in the Ruhr Valley region today. I wrote down this whole log in a real, step-by-step way as I worked—recording all my raw design choices, bad code dead ends, and quick bug fixes right when they happened instead of writing a polished, fake summary at the very end.

The diary is split into clear parts. I have different entries tracking how I came up with my main question, how I hand-digitized the old historic mine shafts and miner colony houses, how I coded the modern walking-distance model, how I calculated the final statistical checks, how I double-checked a totally shocking data flip, how I drew the final charts, and how I later added robustness tests and multi-city data extensions. (Every single number, p-value score, and choice I write down below shows exactly what my computer scripts threw out when I ran the analysis). This includes the crazy final result that completely flipped my starting guess upside down!

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Entry 1

I am calling my project Ghost Infrastructure. It is a mix of old map history and modern walking network code to check if the old 19th and 20th-century coal mines and special worker colony houses (locally called Zechensiedlungen) inside the Ruhr Valley still decide who gets stuck with bad walking paths today, even though the big factory drop started fifty years back and the absolute final mine shut its doors in 1974. At the very beginning, I wanted to map out massive steel factories and old train lines too, but the actual files I collected and processed only cover the basic coal shafts and miner housing. (I chose these specific two because they were the only ones with solid old paper files that I could cleanly read and turn into digital map points). So, instead of lying and saying I finished everything, I wrote down the steel factory and old train line mapping as future work inside my final research paper.

I wanted to change how people think about old factory history. Most geography books just tell long, wordy stories about old factory heritage, but I chose to turn this concept into actual numbers and shapes that a

computer can read and measure. (So, my plan was to take my newly digitized historical maps and drop them straight on top of a modern, network-based walking score to see if it fits the popular 15-minute city layout).

Old habits die hard in city design. Geographers call this path dependency (meaning old layout choices stick around for ages even when the factories close down), but nobody actually proves it with real spatial math. If you read old research about the Ruhr Valley, it is mostly stories about museum heritage, local worker pride, or city planning talks instead of actual calculations to see if old coal layouts change your modern morning walk. At the same time, new papers about the 15-minute city layout look only at modern stuff like your monthly salary, your race, or your age to explain why your neighborhood lacks a good grocery store. (They completely miss looking back at history!). That is the exact blank spot I wanted to fix. I chose to build a clean, repeatable GIS path to turn old paper coal maps into digital coordinate shapes and use network walk models plus spatial math to test if today's unfair walking gaps are actually born from this dead factory geography.

I wrote down one main question. I want to check if the old geography of the Ruhr Valley coal and steel industries—like where the old mines sat, where worker colonies were built, and old train lines—still decides which modern neighborhoods get stuck outside a basic 15-minute walk today.

I set four clear tasks to solve this. First, I chose to digitize and georeference old industrial paper maps for my study zone to catch the old coal shafts and worker housing blocks (I wanted to map old transport lines too but I pushed that to my future work list because it was too much). Second, I chose to build a modern walking network model—generating distance contours to basic daily services like clinics, food shops, primary schools, and public parks—using up-to-date road paths. Third, I chose to apply spatial stats methods like Local Moran's I and hot-spot checks to prove if the worst, low-access areas today are actually grouping up right next to old factory spots instead of just being scattered randomly across the city layout. (Finally, I will generate a direct map overlay of this old history stacked right on top of modern walking scores to visually and mathematically prove if a "ghost infrastructure" shadow is real or completely fake).

I chose Bochum city for this study. It sits right in the North Rhine-Westphalia region of Germany. It was just a small farming town before iron, coal, and steel factories boomed in the mid-19th century, transforming it into a major industrial city through the 1950s (plus it is where my own college, Ruhr University Bochum, is located).

I found excellent old data files online. The official Industrial Heritage Route website (route-industriekultur.ruhr) holds a full list of known historic worker housing colonies within the city borders, giving me a solid list of names to begin my tracing. I set my starting search coordinates between 51.42°N to 51.53°N and 7.13°E to 7.30°E, but I will adjust these limits later when I get the actual old map images.

The upcoming sections show my real work. I am writing down the exact file downloads, data choices, and math check steps I used to test this whole idea. (And yes, this includes a very shocking data result that completely flipped my starting guess upside down!).

Entry 2

I started my next task. I had to build the old history map layer first because all my future work depends 100% on it, so I needed the exact digital latitude and longitude coordinates for all the old Bochum coal pits and worker colony houses.

I collected the data from different web pages. For the old coal mine coordinates, I opened Mindat.org and manually clicked and copied the locations page by page because the website does not have any bulk download button to save time. (Then, for the special miner colony housing fields, I hunted through German

archive pages, local Wikipedia lists, the Route Industriekultur portal, and a building database site called ruhr-bauten.de to find the addresses).

I kept two separate files. My data has exactly 13 coal mines and 4 worker colonies, and I deliberately chose to save them as completely different map layers because a mine shaft and a housing neighborhood are totally different things. While reading my notes again, I spotted a famous steelworker neighborhood called Stahlhausen which was built by the Bochumer Verein factory corporation, but I manually deleted it from my worker colony file because it belongs to the steel business rather than coal mining.

I wrote a quick script. I used GeoPandas to take my text lists and change them into proper GeoPackage map files using the EPSG:4326 coordinate system so I can drop them straight into QGIS software for all my upcoming network tests.

Entry 3

I downloaded the current street map data. I used the OSMnx package to pull the full walking street grid of Bochum directly from OpenStreetMap, which gave me exactly 69,393 node points and 169,668 street line pieces. At the same time, I grabbed 786 daily service spots from the map database—covering local hospitals, medical clinics, medical shops, primary schools, kids' kindergartens, grocery supermarkets, small corner stores, and public green parks.

I built a real network walking path model. (I completely avoided using those lazy straight-line buffer circles because circles pretend you can fly over buildings instead of walking along actual footpaths!). Instead, my script runs Dijkstra's shortest-path algorithm to calculate the actual walking meters along the pavement grid to every single one of these 786 service locations to find out who truly lives a proper 15-minute neighborhood life.

Entry 4

I finally ran my big math check. I had my 17 historical factory shapes ready and my full walking network score completed across 69,393 road points, so I could finally push the button on the exact statistical test I built this entire code pipeline for from day one.

The output completely blew my mind. I ran a Welch's t-test to check the spacing from each road point to the closest old industrial landmark, and the code gave a massive, highly significant relationship score with a t-value of 42.887 and a p-value less than 0.00001! But here is the crazy twist: the arrow was pointing in the completely opposite direction to my original guess. (My data shows that the neighborhoods with the absolute worst walking access are actually sitting much further away from old coal mines at an average distance of 1,984 meters, while the high-access neighborhoods sit super close at just 1,450 meters!). This means living right next to old, dead industrial coal mines actually predicts a much better walkable life today instead of a worse one.

I am reporting this result honestly. I chose to treat this shocking flip as a proper, real finding instead of hiding my face just because my original guess failed completely. See, old factory owners had to build their coal pits right in the middle of thick, packed worker housing areas, and this old, tightly packed city layout still keeps a massive amount of shops and great street links today compared to lonely outer suburbs. (So it is a path dependency of center-spot benefit, not a path dependency of getting ignored!). My first guess was that living near old industrial junk would mean a bad walkable life, but my actual data proves that old factory centrality

gives you a major advantage today, while the real walking gaps are trapped in outer zones far away from the old coal cores.

Entry 5

I needed a major sanity check. Before believing this reversed trend for real, I had to kick out a very obvious hidden trap: old coal mines might just sit right next to Bochum's downtown market center (which naturally has amazing walking paths anyway), so my old factory effect might just be a copycat trick for downtown distance instead of its own independent power.

I ran a quick correlation first. I checked the link between distance-to-factory-sites and distance-to-downtown, and the score came back super tiny at r = 0.063! This proved these two are completely separate map variables, not duplicates of each other. (Then, I coded a logistic regression model to guess my 15-minute neighborhood walk status using both spacing numbers at the exact same time, and guess what? My distance to old industrial pits stayed a super strong independent driver with a coefficient score of -0.0005 and a tiny p-value under 0.001 even after I locked down and controlled for downtown distance). This data check completely proves my "ghost infrastructure" shadow is an actual, honest independent fact and not just a lucky trick of downtown grouping.

Entry 6

I went back to my starting promises. My original project goals explicitly said I must run a Local Moran's I and hot-spot analysis to see if bad walking areas are grouped up together tightly instead of just bouncing around randomly, but my previous t-test and regression equations from Entry 4 and 5 only checked distance metrics between the two walk groups, completely missing the actual map clustering. So I sat down to run the exact math check I promised from day one.

I built a K-nearest-neighbor grid. Setting the neighbor count to 8 across all 69,393 road points, I row-standardized my matrix and calculated the Local Moran's I scores on my binary 15-minute walk flag using 99 random permutations inside Python's `libpysal` and `esda` toolkits (with my random seed locked at 42 and keeping p-value cuts at 0.05).

The math results came out very sharp. My average local I value hit a super high 0.923, and exactly 10,266 out of my 69,393 road points turned out to be part of statistically real spatial groups (which is 14.8% of the whole city network!). Inside this grouped batch, a huge 9,568 pieces were Low-Low cold spots—meaning massive connected patches of bad walking zones surrounded by other bad walking spots—while 698 points came out as weird High-Low spatial anomalies. I found absolutely zero High-High or Low-High clusters anywhere in the results.

(Then I matched these hot-spot groups directly against my old factory spacing columns). I found that road points stuck inside those significant Low-Low cold spots sit at a huge average distance of 1,992.3 meters away from old industrial points, while the ordinary non-significant points sit much closer at just 1,447.1 meters. Also, a massive 97.1% of every single bad-walk point in the entire city is trapped inside an official Low-Low cold-spot group.

My new math style confirms my Entry 4 point completely. I found that bad walking access in Bochum does not just float around randomly, but it forms real, large connected cold-spots on the map that sit measurably farther from old coal industry landmarks compared to the rest of the town. (This directly answers my original task statement exactly). I saved the final cluster image map and my working python code file

`spatial_clustering_lisa.py` straight into the main repository folder, and then I typed down all the exact math scores inside section 4.4 of my `GI_Research_Paper.md` file.

Entry 7

My map code broke completely on the first try. I wanted to generate my project's main image visualization—stacking all 69,393 road points colored by walking scores right next to my old coal shaft and colony points—but my screen came out totally blank showing only one single tiny dot in the corner! I sat down and investigated this mess instead of panic-clicking. (I found out it was a classic coordinate projection mismatch bug). See, I had saved my modern street network file in EPSG:32632 UTM Zone 32N because I needed a metric scale for running my distance math earlier, but my old history files were still stuck in regular EPSG:4326 latitude and longitude degrees.

Drawing these two different types on the same canvas without fixing the projection was a total disaster. The metric UTM map coordinates use massive numbers in the hundreds of thousands of meters, which forced the tiny latitude-longitude coordinates to shrink into an invisible single dot on the screen. I manually reprojected every single layer into the exact same EPSG:4326 grid before running my matplotlib code, and the whole city map popped up beautifully.

My map visual had two strange glitches. When I checked my map layers closely, I spotted only 12 triangle symbols instead of my 13 coal mines, and 2 of my 4 worker colony blocks looked completely mashed on top of each other. I did not assume my code script was broken, so I opened my raw data coordinate spreadsheet rows to check the actual numbers behind them.

The numbers proved my map was completely right. See, the Mansfeld and Heinrich Gustav coal mines in the Langendreer and Werne areas sit only 1.7 kilometers away from each other, which forces them to completely merge into a single dot at a full-city map zoom level. (Similarly, two separate worker neighborhoods named Kolonie Hannover and Am Rübenkamp sit well under a kilometer apart because factory heads built both next to each other to feed the exact same Hannover mine shafts during overlapping construction phases between 1874 and 1892). I did not have to alter a single line of my code pipeline because these overlapping spots were real, historical clustering on the ground rather than messy database typing errors.

This whole exercise taught me a solid rule for all my future coding work. When my data threw out a totally unexpected flipped result, I did not just ignore it or assume it was a code glitch. Instead, I immediately ran a real correlation and a multiple regression model back in Entry 5 to prove it was an independent fact rather than some hidden trick. (And when my map visual looked strange, I did not just hit delete; I went back to the raw spreadsheet values to check the history). I forced myself to track every weird glitch down to its real cause before making any final claims instead of faking the story or accepting it blindly.

But people kept pointing out the matching dots on my map. My review notes kept highlighting the overlapping colony boxes and the missing 13th mine symbol, so I decided to calculate the exact Haversine distance equations between the overlapping coordinate points rather than just staring at the map frame by eye.

The math gave me the exact numbers. My script showed that the Mansfeld and Heinrich Gustav coal mines are precisely 1.75 kilometers apart from each other, while the Kolonie Hannover and Am Rübenkamp worker neighborhoods sit at a tiny distance of just 0.33 kilometers. Since the full city of Bochum spans a massive 14 kilometers across the map canvas, these tight gaps naturally force the symbols to merge into a single marker point at a full-city scale. This completely closes my verification loophole. My raw base data is officially checked

out and true because I have all 13 separate mines with valid coordinates sitting in my sheet, and now my distance loop proves that the overlapping icons are real historical geography rather than lazy data entry or pipeline bugs.

Entry 8

I finished all my coding work. Since my main calculations, confound checks, hot-spot maps, and visual charts were all completely done, my last major task was to put everything together nicely inside a multi-page Streamlit web app dashboard.

I wrote the master code setup file named `app.py` to act as the main front door for the website. I split the dashboard into separate sub-pages covering my starting study plans, old factory maps, walking accessibility charts, my main reversed finding text, trend lines, live interactive maps, and my exact data methodology steps. (Doing this lets anyone open the dashboard and play around with my historical overlays and statistical numbers directly inside an interactive internet browser instead of just reading a long, boring, flat text document!). Along with this live dashboard app, I drafted my final Research Paper, completed the full Project Report file, typed up a clean `README.md` instruction file, and pushed my entire code repository straight onto GitHub for my final submission.

Entry 9

I am expanding all my finished portfolio projects now. I want to add missing files, stretch out the target scope, and wipe out my documented limitations wherever possible, so I selected this Bochum project as my absolute first candidate because it needs zero Google Earth Engine tools (it just runs on pure OSMnx, OpenStreetMap data, and open GADM borders) and it already has a handy "Future Work" checklist written down.

I started with a zero-new-data task. I chose to code a time-limit sensitivity check because my old paper noted it as an incomplete item, so I wrote a brand new file named `threshold_sensitivity.py` in the main root folder to rerun the full walk-classification, distance loops, Welch's t-tests, and logistic regression confound formulas using different 10-minute (750 meters) and 20-minute (1,500 meters) boundaries. (I recycled the exact same Bochum street grid and distance columns I previously calculated, so I did not have to download any fresh map layers).

My reversed finding survived everywhere. In fact, the link gets even more powerful as you expand the walk limits! For the 10-minute walk, I calculated a t-score of 47.062 with a p-value less than 0.00001 and a Cohen's d of 0.413, while my old 15-minute run sat at t = 42.887 and d = 0.589, and my new 20-minute run hit a t-score of 32.150 with a d-value of 0.661. My odds ratio score for getting 100 meters closer to an old mine remained super stable across all three options at 4.24%, 4.88%, and 4.49%, which proves my original 15-minute result was a real geographic fact and not just a fluke from a single chosen walk limit. I pasted this full breakdown straight into Section 4.5 of my `GI_Research_Paper.md` file.

Entry 10

I picked Essen for my second city test. My own paper's Future Work list explicitly stated that I must repeat this exact code logic in other matching Ruhr Valley towns, so I chose Essen because it sits just 15 kilometers northeast of Bochum, shares the exact same 19th-century deep coal mining history, and has way better local heritage files than any other alternative place.

But my web data collection hit a massive server wall very early on. OpenStreetMap's main download lines like the Overpass API and Nominatim were completely down during my data gathering phase (which is the exact same annoying connection blocker I faced on my other Earth Engine projects). To make things worse, Wikidata's live links failed too, and Mindat.org—the exact site I used to build my first Bochum mine list—threw an ugly 403 forbidden error when my script tried to scrape it!

(I managed to find a very clever workaround to save my data). I found that KuLaDig, which is the official North Rhine-Westphalia state cultural heritage map database, was working perfectly fine and gave me exact WGS84 map coordinates for all the main heritage-listed mine shafts. At the same time, I noticed that German Wikipedia articles for worker housing neighborhoods always carry clean, geo-tagged coordinate boxes right inside the text page, which is great because pages for the blown-up mine shafts usually leave those numbers out completely.

I hand-digitized four massive coal mines using this trick. I caught Zeche Zollverein Schacht XII in Katernberg (the big UNESCO World Heritage spot — the shaft itself ran 1932-1986, part of the wider Zollverein colliery complex founded 1847/51), Zeche Carl Funke down in Heisingen (1804-1973), Zeche Vereinigte Helene & Amalie in Altendorf (1843/44-1968), and Zeche Pörtingsiepen in Fischlaken (1779-1972). Then I also grabbed four main worker housing colony neighborhoods: Siedlung Carl Funke built in 1900, Mathias-Stinnes-Siedlung in Karnap from 1890, Kolonie Zollverein III around Katernberg (1880), and Kolonie Beisen from 1902. (Before doing any script work, I locally clipped the Germany GADM v4.1 boundary map file that was already sitting in my laptop folders to double-check that all 8 points fell perfectly inside Essen's real border lines as a basic mapping sanity test).

This is a much smaller list than my 17 Bochum points. See, Essen's official history website claims there are over 1,700 total old mining spots across the city, so my small list is just a highly precise subset of the major landmarks, and I am listing this honestly as a project limitation rather than hyping it up as a total census check. Funny thing is, I originally started with a tiny 3-mine and 3-colony setup (just 6 points total!). But when my Entry 11 statistics checks came out a bit fuzzy, I realized my small sample size was messing with the code, so I manually went back and added two more points to grow my file to the current 8-site version.

I downloaded the Essen walking street layout. I wrote a brand new download script named `download_network_essen.py` that mirrors my old Bochum code exactly, but I had to point it directly at the Overpass API server lines to fetch Essen's full pavement map layer. Running it locally on my laptop pulled a huge grid with 72,027 road intersection nodes and 188,198 street edge line segments, along with a total of 1,410 raw city facility points.

(Then I had to clean up the messy data). I applied the exact same strict point-geometry filtering code I built for my first Bochum analysis, which chopped down that massive list to exactly 366 highly usable daily service locations.

Entry 11

I ran my fresh Essen script. I built a brand new code pipeline file called `run_essen_pipeline.py` which replicates my original step-by-step math from Entries 3 through 6 for this new city, keeping every single setting and code argument exactly identical to my Bochum code (using that same 1,125 meters walk limit, a Welch's t-test calculation, Essen Central Station as the city center coordinate, and my KNN 8-neighbor cluster check with 99 permutations on a locked 42 random seed).

The final percentages came out surprisingly close. My script showed that 88.5% of Essen's 72,027 total street points sit safely inside a 15-minute neighborhood walk (which is slightly better than Bochum's 85.8% score). (The main reversed finding popped up beautifully a second time!). My data shows that the 8,267 bad-walk points sit at a huge average distance of 3,693 meters away from the nearest old coal mine, while the 63,760 high-access points sit much closer at an average of 3,130 meters. This gave a Welch's t-score of 24.731 with a p-value less than 0.00001 and a Cohen's d value of 0.338. (This effect size is smaller than Bochum's 0.589 score, but the arrow points the exact same way and stays highly significant!).

My Local Moran's I cluster calculations also matched up nicely across both places. Essen's average local I value hit 0.917 compared to Bochum's 0.923, and a massive 95.5% of every bad walking point in Essen fell inside a statistically verified Low-Low cold-spot cluster. Just like my first city check, I found absolutely zero significant High-High hot spots anywhere on the map.

But my center-spot checking code hit a massive snag here. The link between mine distance and downtown distance came out at a much higher r = 0.405 in Essen compared to that tiny r = 0.063 back in Bochum (so this is a totally unique geographic difference, not just random math noise). When I ran the controlled logistic regression model, the old factory coefficient number completely flipped its sign over to +0.0001 with a p-value under 0.00001 compared to Bochum's old score of -0.0005! This means that once you freeze and control for downtown distance, living farther away from an old mine actually links to a higher chance of good walking paths inside Essen.

I recalculated this twice to be super sure. (My starting 6-site data layer gave a higher r-score of 0.475 with that same flipped coefficient model). I intentionally added two more sites to grow my file to 8 points, which successfully dragged the correlation down to r = 0.405, but it completely failed to stop that annoying sign flip from happening. This proves that while my low sample size was partly messing up the numbers, it cannot explain away the whole sign flip story by itself.

Entry 12

I stuck to my old data rules from Entry 7. When Essen threw out that weird confound-check mismatch, I did not try to hide it or run away from it like a scared student. I wrote down and tested two logical reasons for this gap instead of choosing whichever looked easier for my story.

First, the two cities might simply have totally different maps on the ground because Essen's old coal mines might be huddled much closer to its modern downtown compared to Bochum's scattered locations. Second, it could just be a silly data data error because my Essen history file has very few points compared to my Bochum file (8 versus 17).

The math gave me a very clear clue. My measured correlation score dropped from r = 0.475 when I used 6 sites down to r = 0.405 when I grew it to 8 sites, which proves my small sample size is definitely messing up the regression line a little bit. (But this data trend alone cannot fully disprove the first map reason!). So, instead of forcing a fake answer to look smart, I wrote down both possibilities and printed the exact numbers for each inside sections 4.6 and 7 of my `GI_Research_Paper.md` file. I officially labeled expanding the Essen data sheet as future work because finding precise map coordinates for their remaining, poorly-documented old factories was getting nearly impossible this week.

I drew one big conclusion from this multi-city extension. The main center-spot benefit idea—where the raw flipped trend and its separate map clustering match up—seems to hold true across both of these Ruhr Valley towns. But my smaller, extra claim that this old mine advantage works completely independently of downtown

proximity looks like a unique Bochum-only fact rather than a universal rule based on my current two-city maps. (I chose to write down this messy, mixed result in full detail instead of hiding the parts that did not match my first city's findings perfectly). This felt like a much more useful and honest choice to make, matching exactly how I handled every single other shocking data twist in this project diary.

Entry 13

I noticed a bad mismatch in my map layouts. My old Bochum overlay map was still using that outdated QGIS2Web software template from the early weeks of my study, while my new Essen map was already coded directly in Python using Folium libraries. This was a super annoying gap to leave behind because both of my study cities should be treated exactly the same on paper.

I completely rebuilt the file `outputs/maps/bochum_interactive_map.html` from scratch using my local geopackage vector files (loading my calculated Bochum walking metrics, coal shaft points, Zechensiedlungen coordinates, and clipping my GADM administrative boundary shape). I styled it to look like a twin copy of my Essen layout. I picked dark CartoDB background tiles, dropped a bright orange industry-icon marker on every coal mine, and stuck a light blue home-icon marker on every single worker neighborhood. (If a user clicks on any pin, a little popup box shows the factory name, the local neighborhood district, and the working history dates!).

Next, I handled my huge point count. I had a massive 9,858 low-walk road points, so instead of drawing individual markers that would fully freeze the internet browser (just like Essen's 8,267 nodes would have done), I passed them through a smooth heatmap density layer instead. Lastly, I styled the outer city border lines with the exact same teal and cream layout color scheme I chose for Essen.

My Bochum popups have one small difference from Essen. My Essen pins show a direct web source link because I collected KuLaDig and Wikipedia URLs during that step, but my raw Bochum files do not have a citation column saved in the spreadsheet. (So the Bochum popups only print out the name, the area district, and the working years). I did not change any coordinate positions or operation dates during this full rewrite.

The main database stayed completely untouched. Every single starting file for Bochum—like my hand-digitized coal pits, the main 69,393-point network grid, and the outer border shape—uses the exact same inputs that my old QGIS software read weeks ago. This task was a 100% code rendering clean up. I successfully threw out that heavy, multi-file QGIS2Web export folder and replaced it with a single, lightweight HTML file built directly from my own geopackages.

I updated my dashboard code files next. I opened `dashboard/pages/6_Interactive_Maps.py` to embed the fresh Bochum map file using `components.html` syntax so it matches my Essen page structure perfectly instead of calling that old broken iframe folder path. Then, I adjusted the text for the map legends and edited my main `README.md` file to delete all mentions of the old QGIS2Web software tools. Now, both cities have their web maps built the exact same way from scratch using the same visual look, which means I have zero QGIS dependencies left in my map pipeline.